

MANIPAL ACADEMY OF HIGHER EDUCATION, MANIPAL
MANIPAL INSTITUTE OF TECHNOLOGY
MID-SEMESTER RETEST EXAMINATION, APRIL 2025
MAT 2226 - ENGINEERING MATHEMATICS - IV

Date: 17-14-2025

Time: 5:00-5:20 PM

Marks: 05

1. If $\mu = 3$ and $\sigma = \frac{1}{\sqrt{3}}$, then $P(|X - 3| \geq 1) \leq \dots$
 - a. $\frac{1}{3}$
 - b. $\frac{1}{\sqrt{3}}$
 - c. $\frac{1}{2}$
 - d. $\frac{2}{\sqrt{3}}$
2. If X is a Poisson variate such that $P(X = 0) = \frac{1}{2}$, then $V(X) = \dots$
 - a. 0.5
 - b. $\ln(2)$
 - c. 2
 - d. 0.25
3. Two events A and B satisfy $P(A) = 0.2$, $P(B) = 1$, $P(A | B) = 0.4$. Which of the following is true?
 - a. A and B are independent.
 - b. A and B are mutually exclusive.
 - c. $P(A \cap B) = 0.2$
 - d. $P(A \cap B) = 0.4$
4. A and B are independent events. The odds in favour of A and B are 1:1 and 3:2 respectively. Find the probability that only one of the two events occur?
 - a. 0.6
 - b. 0.7
 - c. 0.8
 - d. 0.5
5. A binomial distribution has parameters $n = 10$ and $p = 0.4$. What is the probability of getting exactly 4 successes?
 - a. 0.2508
 - b. 0.1115
 - c. 0.2890
 - d. 0.4000

6. If X and Y are two random variables having joint probability distribution function $f(x, y) = \frac{6-x-y}{8}$, $0 < x < 2$, $2 < y < 4$, Then the probability $P(Y < 3)$ is...
- a. $3/8$
 - b. $5/8$
 - c. $5/3$
 - d. $3/5$
7. Suppose X and Y are random variables such that $V(X) = 1$, $V(Y) = 3$ and $Cov(X, Y) = 2$. Then the covariance of $2X + Y$ and $X - 2Y$ is...
- a. 2
 - b. 3
 - c. -8
 - d. -10
8. A standard deck of 52 cards is shuffled. Cards are drawn one by one without replacement until the first heart appears. The probability that the first heart appears on the 3rd draw is approximately...
- a. 0.0968
 - b. 0.1453
 - c. 0.2123
 - d. 0.2304
9. A man tosses a coin and throws a die alternatively, beginning with coin. What is the probability that he will get head before he gets 5 or 6 on the die?
- a. $\frac{1}{2}$
 - b. $\frac{3}{4}$
 - c. $\frac{4}{5}$
 - d. $\frac{1}{4}$
10. A continuous random variable has pdf $f(x) = kx^2e^{-x}$; $x \geq 0$. Then the value of $k = \dots$
- a. 3
 - b. $\frac{1}{2}$
 - c. $\frac{1}{4}$
 - d. 2